29. If X is a Poisson variate with $P(X=1) = P(X = 2)$, find its mean.

30. Why U (a, b) is called a rectangular distribution?

31. If $X \sim$ U (-10, 10) find its mean and variance.

32. State the conditions for B (n, p) to become N (μ, σ^2)

33. State the limiting conditions for a Poisson distribution to become a Normal distribution.

34. Evaluate the value of P ($|X - \mu| < 3\sigma$), when $X \sim N(\mu, \sigma^2)$

35. Give a note about Standard Normal distribution.

36. State the properties concerned with the area of a Normal distribution.

IV. Answer in brief:

37. 10% of the screws manufactured by an automatic machine are found to be defective. 20 screws are selected at random find the probability that (i) exactly 2 are defective (ii) at most 3 are defective (iii)at least 2 are defective.

38. 20 wrist watches in a box of 100 are defective. If 10 watches are selected at random find the probability that (i) 10 are defective (ii) at most 3 are defective.

39. A manufacturer of television sets knows that an average of 5% of the product is defective. He sells television sets in consignment of 100 and guarantees that not more than 4 sets will be defective. What is the probability that a television set will fail to meet the guaranteed quality? [e-5 = 0.0067]

40. A factory employing a huge number of workers find that over a period of time, average absentee rate is three workers per shift. Calculate the probability that in a given shift (i) exactly 2 workers (ii) more than 4 workers will be absent?

41. The remuneration paid to 100 lecturers coaching for professional entrance examination are normally distributed with mean Rs. 700/- and standard deviation Rs. 50/-. Estimate the number of lectures whose remuneration will be (i) between Rs. 700/- and Rs. 720/- (ii) more than Rs. 750/-.

V. Calculate the following:

42. The skewness and kurtosis of a binomial distribution are 1/6 and – 11/36 respectively. Find the Binomial distribution.